

Question ID: 33e29881

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question

In right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. The value of $\sin(R)$ is $\frac{\sqrt{15}}{4}$. What is the value of $\cos(S)$?

Answer

- A. $\frac{\sqrt{15}}{15}$
- B. $\frac{\sqrt{15}}{4}$
- C. $\frac{4\sqrt{15}}{15}$
- D. $\sqrt{15}$

Correct Answer: B

Rationale

Choice B is correct. The sine of any acute angle is equal to the cosine of its complement. It's given that in right triangle RST , the sum of the measures of angle R and angle S is 90 degrees. Therefore, angle R and angle S are complementary, and the value of $\sin R$ is equal to the value of $\cos S$. It's given that the value of $\sin R$ is $\frac{\sqrt{15}}{4}$, so the value of $\cos S$ is also $\frac{\sqrt{15}}{4}$.

Choice A is incorrect. This is the value of $\tan S$.

Choice C is incorrect. This is the value of $\frac{1}{\cos S}$.

Choice D is incorrect. This is the value of $\frac{1}{\tan S}$.